The objective of this task was to understand the basic functionality of a firewall, configure firewall rules, block network traffic on a specific port, test the rule, and restore the original firewall configuration.

TASK 4: SETUP AND USE A FIREWALL ON WINDOWS

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Introduction

A firewall is a network security mechanism that monitors and filters incoming and outgoing network traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks.

Firewalls help protect systems from unauthorized access, malicious traffic, and various cyber threats.

Tools Used

- Windows 11 Pro
- Windows Defender Firewall with Advanced Security
- Windows PowerShell

Procedure

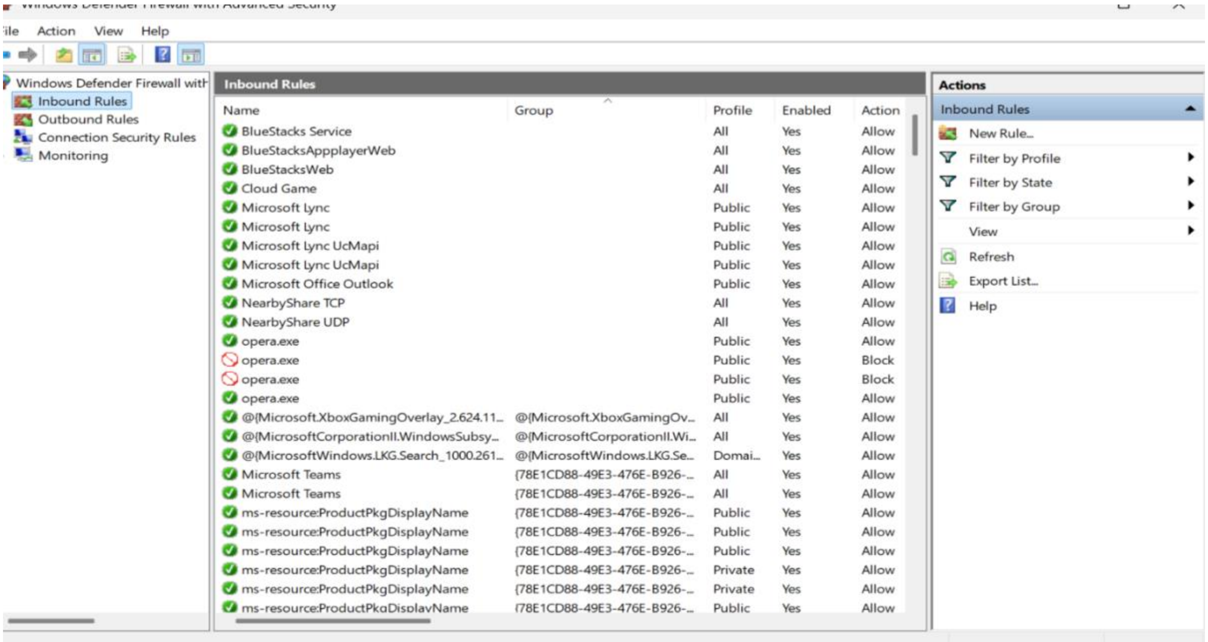
Step 1: Open Windows Firewall

Windows Defender Firewall with Advanced Security was opened from the Windows Search menu.

View Existing Firewall Rules

The existing inbound firewall rules were reviewed to understand the current firewall configuration.

Screenshot 1:



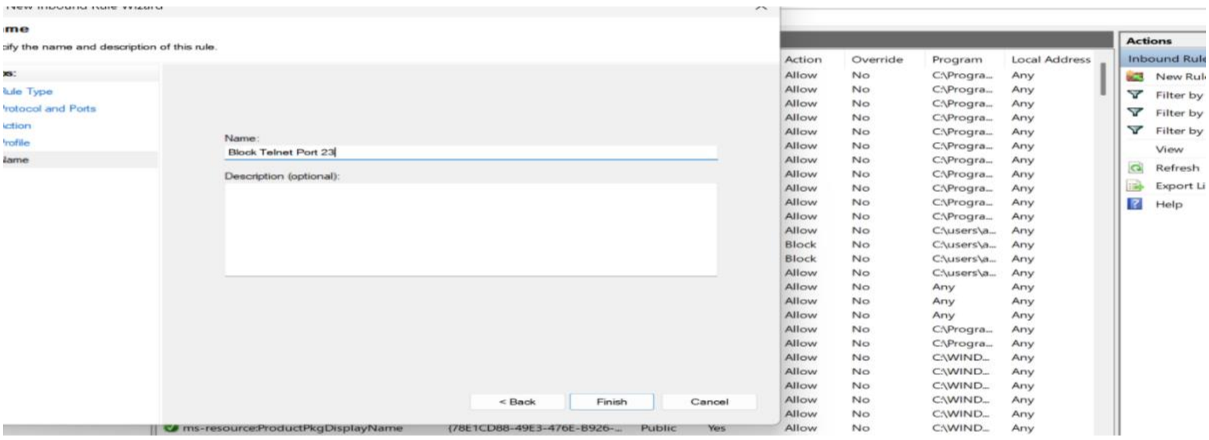
Create a New Firewall Rule

A new inbound firewall rule was created using the following configuration:

- Rule Type: Port
- Protocol: TCP
- Port Number: 23
- Action: Block the Connection
- Profile: Domain, Private, Public
- Rule Name: Block Telnet Port 23

The purpose of this rule was to block inbound traffic on Port 23.

Screenshot 2:



Verify the Rule

The newly created firewall rule was verified in the Inbound Rules section to ensure it was successfully applied.

Test the Firewall Rule

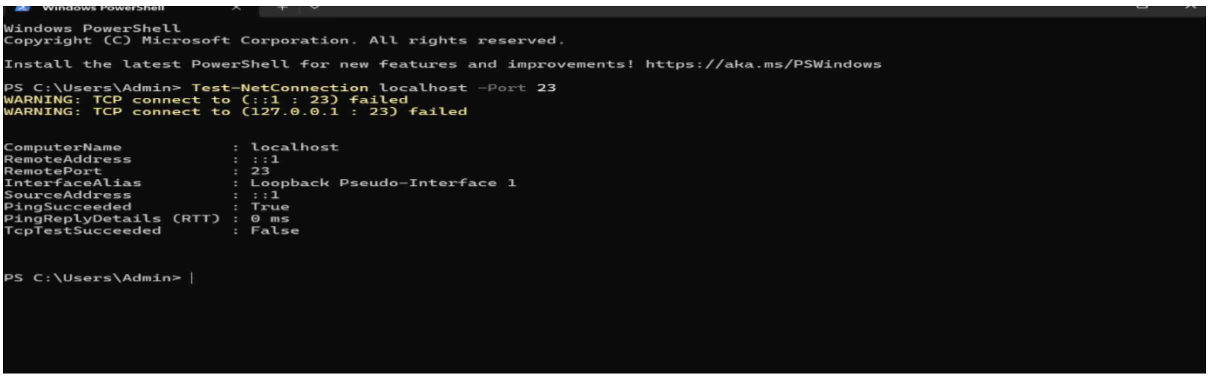
The firewall rule was tested using Windows PowerShell.

Command Used:

Test-Net Connection localhost -Port 23

The test confirmed that the firewall rule was active and capable of controlling traffic on the specified port.

Screenshot 3:

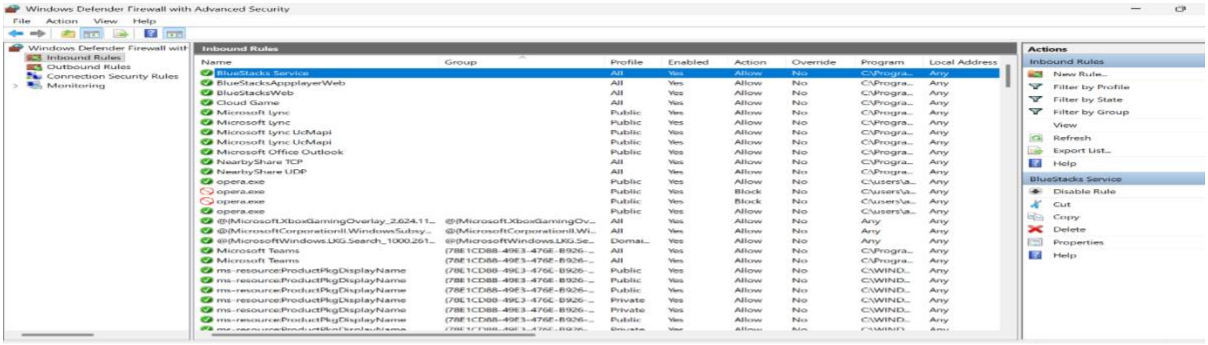


Remove the Test Rule

After verification and testing, the firewall rule was removed to restore the original firewall configuration.

This follows security best practices by removing temporary testing rules after use.

Screenshot 4:



Results

The Windows Firewall was successfully configured to block inbound traffic on Port 23 (Telnet). The rule was verified and tested successfully. After testing, the temporary rule was removed to restore the system to its original configuration.

Understanding Port 23 (Telnet)

Port 23 is used by the Telnet protocol.

Telnet is considered insecure because it transmits usernames and passwords in plain text without encryption. Due to this security risk, Telnet is rarely used in modern environments and is often blocked by firewalls.

Firewall Traffic Filtering

A firewall filters network traffic based on predefined security policies. It examines packets and determines whether they should be allowed or blocked according to configured rules.

Firewall filtering helps:

- Prevent unauthorized access
- Reduce attack surface
- Control network communication
- Improve overall system security

Interview Questions and Answers

1. What is a firewall?

A firewall is a security system that monitors and filters incoming and outgoing network traffic based on predefined security rules.

2. What is the difference between a stateful and stateless firewall?

Stateful firewalls track active connections and make decisions based on connection state. Stateless firewalls inspect each packet independently without maintaining connection information.

3. What are inbound and outbound rules?

Inbound rules control traffic entering a system, while outbound rules control traffic leaving a system.

4. How does UFW simplify firewall management?

UFW provides a user-friendly interface for managing firewall rules on Linux systems and simplifies complex firewall configurations.

5. Why is Port 23 commonly blocked?

Port 23 is used by Telnet, which sends data without encryption, making it vulnerable to interception and credential theft.

. What are common firewall mistakes?

Common mistakes include allowing unnecessary ports, failing to update rules, ignoring firewall logs, and not removing temporary rules.

7. How does a firewall improve network security?

A firewall blocks unauthorized access, filters malicious traffic, and enforces security policies to protect systems and networks.

8. What is NAT in firewalls?

NAT (Network Address Translation) translates private IP addresses into public IP addresses and helps conserve IP addresses while providing additional security.

Conclusion

In this task, Windows Defender Firewall was used to create, test, and remove a firewall rule. The exercise provided practical experience with firewall configuration, traffic filtering, and basic network security concepts. The task improved understanding of how firewalls protect systems from unauthorized access and cyber threats.